Tutorial 3: Method of moments and maximum likelihood estimation

1. For random (i.i.d.) samples of size n from the following distributions, find the MLE and the MME of the indicated parameters:

   (a) p in Geo (p) with probability mass function given by $f(x;p) = p(1-p)^{x-1}$ for $x = 1, 2, 3, ...$

   (b) λ in Gamma (t, λ) with $t > 0$ known.

   (c) θ in Beta$(\theta, 1)$, sample size n. (Density is given by $f(x;\theta) = \theta x^{\theta-1}$ for $0 < x < 1$).

2. A health-and-safety officer believes that the number of industrial accidents occurring each month comes from a Poi(λ) distribution with the numbers being independent between months. In 3 successive months they observe 3, 6, and 2 accidents, respectively.

   (a) Construct the likelihood $L(\lambda)$ and the loglikelihood $l(\lambda)$ for this data set.

   (b) Maximise the log-likelihood to obtain the MLE of λ for this data set.

   (c) Find the method of moments estimator for λ. How does it compare with the MLE?

3. The lifetime of a certain type of lightbulb is distributed as an exponential random variable with parameter λ, i.e. the mean is $1/\lambda$. A sample of n bulbs is put on test and testing is stopped after time t_0, by which time x of the n bulbs have failed. Let X be the random variable corresponding to the observed value x.

   (a) Explain why $X \sim \text{Bin}(n, 1 - \exp(-\lambda t_0))$.

   (b) Hence, find the method of moments estimator (MME) of λ based on the distribution of X in (a).

   (c) Find the maximum likelihood estimator (MLE) of λ, again based on (a).

4. Suppose that $\underline{X} = (X_1, X_2, ..., X_n)$ is a random sample from an Exp$(1/\theta)$ distribution with mean $\theta > 0$. Let

$$X_{(1)} \equiv \min\{X_1, X_2, ..., X_n\}.$$

   (a) Determine the distribution of $X_{(1)}$ and hence, show that $g(\underline{X}) \equiv nX_{(1)}$ is an unbiased estimator for θ.

   Hint: To determine the distribution of $X_{(1)}$, consider $\mathbb{P}\{X_{(1)} > x\}$ for $x > 0$. (Recall that $\{X_{(1)} > x\} = \{X_1 > x, X_2 > x, ..., X_n > x\}$.)

   Fact (from earlier Stats courses): If $Y \sim \text{Exp}(\lambda)$ and $c > 0$, then $cY \sim \text{Exp}(\lambda/c)$.

   (b) Calculate MSE$(g(\underline{X}))$ where $g(\underline{X}) = nX_{(1)}$.

5. Suppose that $\underline{X} = (X_1, X_2, ..., X_n)$ is a random sample from a distribution with density function

$$f_X(x;\theta) = \begin{cases} \frac{2x}{\theta^2} & \text{if} \quad 0 \leq x \leq \theta, \\ 0 & \text{if } x < 0 \text{ or } x > \theta. \end{cases}$$

   (a) Suppose $\underline{x} = (x_1, x_2, ..., x_n)$ is a realisation of $(X_1, X_2, ..., X_n)$.
   Determine the likelihood function $L(\theta; \underline{x})$ and sketch its graph.
   Use the graph of $L(\theta; \underline{x})$ to show that the maximum likelihood estimate of θ is equal to $x_{(n)} = \max\{x_1, x_2, ..., x_n\}$.

   (b) Let $X_{(n)} \equiv \max\{X_1, X_2, ..., X_n\}$.
   Determine the probability density function for $X_{(n)}$ and calculate $\mathbb{E}(X_{(n)})$ and $\mathbb{V}\text{ar}(X_{(n)})$.
   Hint: Work in a similar way as in Q4 for $X_{(1)}$.

   (c) Determine the constant γ such that $\gamma X_{(n)}$ is an *unbiased* estimator of θ.
   For this value of γ compare the efficiency of the unbiased estimators $\gamma X_{(n)}$ and $\hat{\theta} = \frac{3\bar{X}}{2}$ (see Question 8, Tutorial Sheet 2).

6. A random sample of 10 policies from a health insurance portfolio, gave the following numbers of *claims per policy*:

   `1 0 2 0 1 2 1 0 0 1`

   It is believed that the number of claims per policy is a random variable X with $X \sim \text{Poi}(\lambda)$. Find the ML estimate of the probability that a policy from this portfolio will give 2 or more claims.

   (*Hint: find the MLE of λ, express the probability of interest as a function of λ and then use the invariance property of MLE.*)

7. The time till relapse, in months, of patients treated with a certain drug is believed to follow a Gamma$(2, \beta)$ distribution where β is unknown. To estimate β you take a random sample of 10 patients at time $x = 0.0$ and monitor them over the following 6 months. In this time 6 patients relapse at times $x_1, \ldots, x_6$ for which $\sum x_i = 13.2$. The remaining 4 patients do not relapse (i.e. times are *censored*).

   (a) Show that the probability that a given patient does not relapse during the monitoring period is given as a function of (β) by

   $$p(\beta) = (1 + 6\beta)e^{-6\beta}.$$

   (*Hint: use integration by parts.*)

   (b) Explain carefully why the likelihood satisfies

   $$L(\beta) \propto \beta^{12}(1 + 6\beta)^4 e^{-37.2\beta}.$$

   Hint: You need to include a factor corresponding to each of the 10 patients.

   (c) Find a quadratic equation satisfied by the MLE of β and hence find the MLE. (*Remember that β must be positive!*)

Exam Style Question

Consider a random sample of size n, $\underline{X} = (X_1, \ldots, X_n)$ from a distribution having probability density function

$$f(x;\theta) = e^{\theta - x} \quad \text{for } \theta \leq x < \infty.$$

(a) Find the method of moments estimator (MME) of θ.

[5 marks]

(b) Give an equation for both the likelihood function $L(\theta; x)$ and the log-likelihood function $\ell(\theta; x)$.

[2 marks]

(c) Show that the maximum likelihood estimator (MLE) of θ is

$$\hat{\theta}(\underline{x}) = \min(x_1, \ldots, x_n).$$

[2 marks]

[Total 9 marks]